

Diagnosing Parkinson's By Applying Graph Neural Networks to MRI

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100% Presentation; Syslab, Yilmaz, Pd 7



PROBLEM

- Parkinson's disease (PD) is a predominately clinical diagnosis, based on identification of characteristic motor symptoms [9]
- [16] showed that diagnoses have accuracy of 80.6%
- [4] found that diagnoses of PD expects 20% error rate



BACKGROUND

- Parkinson's is a disorder that results in degradation of dopamine-producing neurons
- PD also associated with cortical thinning - e.g., reduced prefrontal cortex
 - Can be detected before patients have severe mental symptoms, e.g. dementia [20]



BACKGROUND

- MRI can be converted to cortical surfaces via FreeSurfer [8]
- Surfaces consist of meshes w/ vertices
- Each vertex has various features (curvature, sulcal depth, etc.)
[8]



OTHER SOLUTIONS

- Past studies applying ML to diagnosing PD have focused on volumetrics (e.g., 3d CNNs) [7] Maximum of 95.3% Testing Accuracy on 5-split cross-validation
 - Used very limited dataset (406 individuals) w/ extremely specific conditions (specific angle, MPRAGE manufacturer, field strength)
 - Only considered control & Parkinson's Disease patients

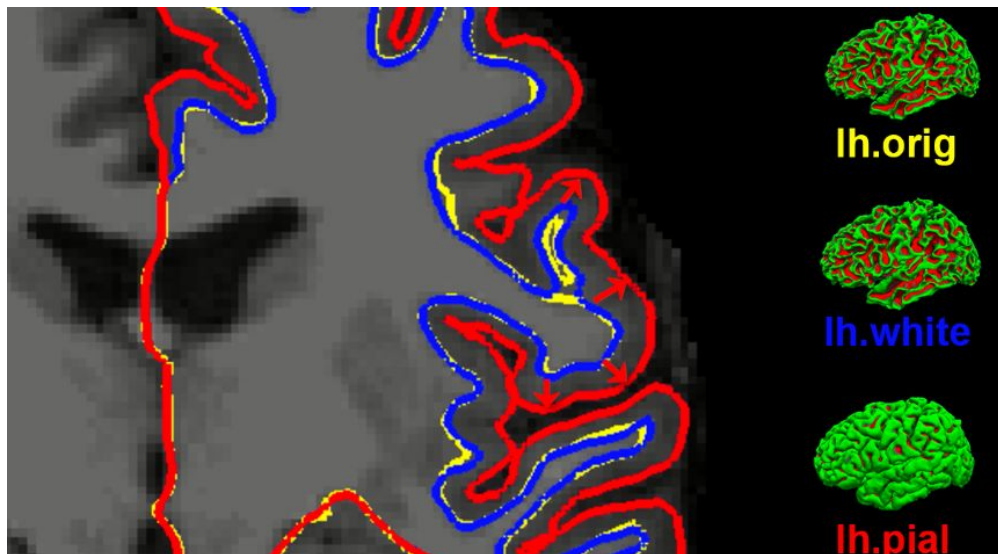


OTHER SOLUTIONS

- Also used DaTSCAN (measure of dopaminergic connection) w/ CNNs [14] 99.9% Testing accuracy on best-performing model
 - While study with DaTSCAN show higher accuracy, DaTSCAN is a very expensive/time-consuming procedure
 - MRI is often easier/more accessible for most patients
 - Also differentiates only between control and PD

NOVELTY

- Apply graph-based neural network to cortical surfaces to detect Parkinson's Disease





WHY IS OURS BETTER?

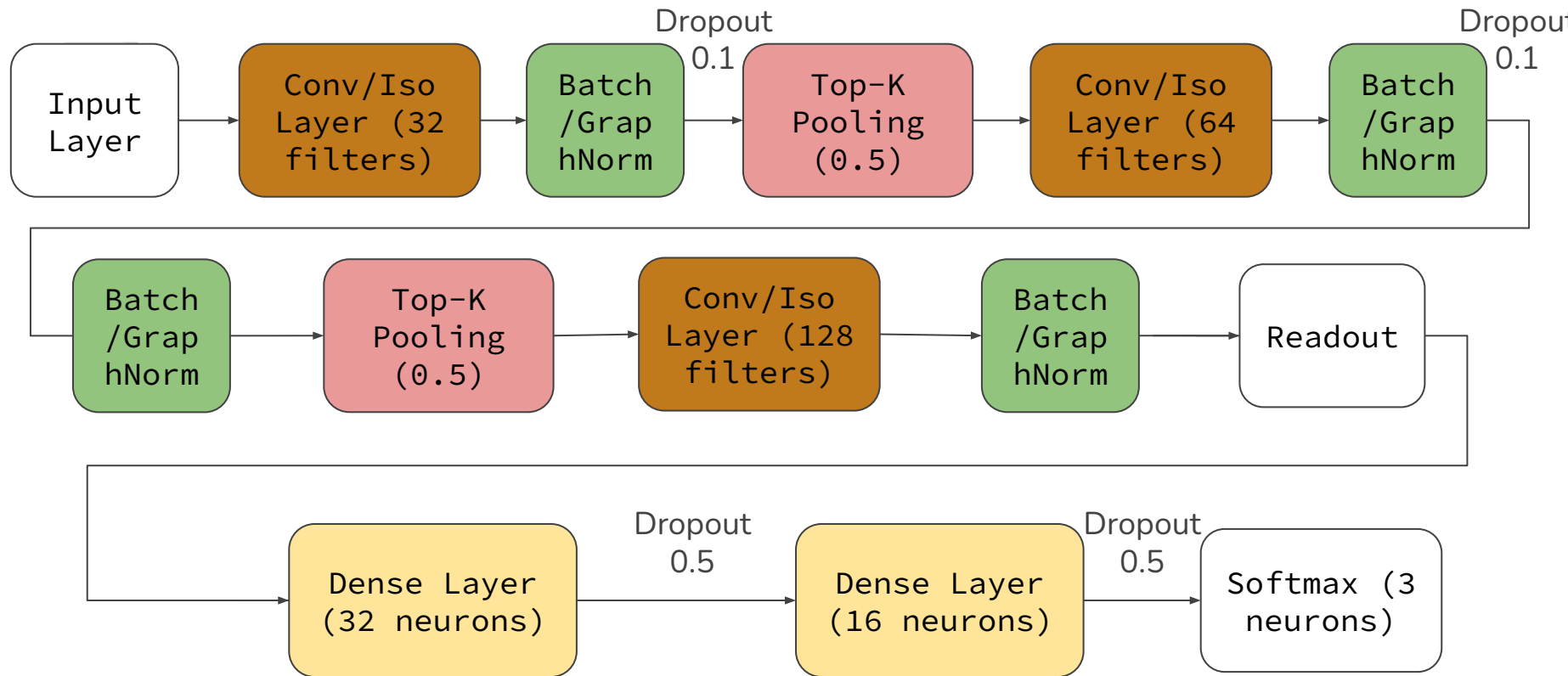
- GNN applied to cortical surfaces picks up directly on features of thickness, sulcal depth, etc. that are noted to decline in PD
- Also, more efficient than similar 3D-CNNs, as those rely on 3D weight matrix over volume of the surface
 - GCN flattens surface into graph with 2D weight matrix over points on surface (e.g. nodes) -> uses less memory/faster training
- Apply to fairly general case (~3000 samples across 3 different degrees of PD)



IMPACT

- Correct diagnoses earlier -> better management
 - Could potentially determine subtle signs (e.g. Prodromal) in patients years earlier (4-8 years) [29]
- Shown that younger age at diagnosis leads to longer survival [17]
- PD is treated with deep brain stimulation to reduce tremors [13]
- Also treated w/ chemical agents that mimic dopamine [15]

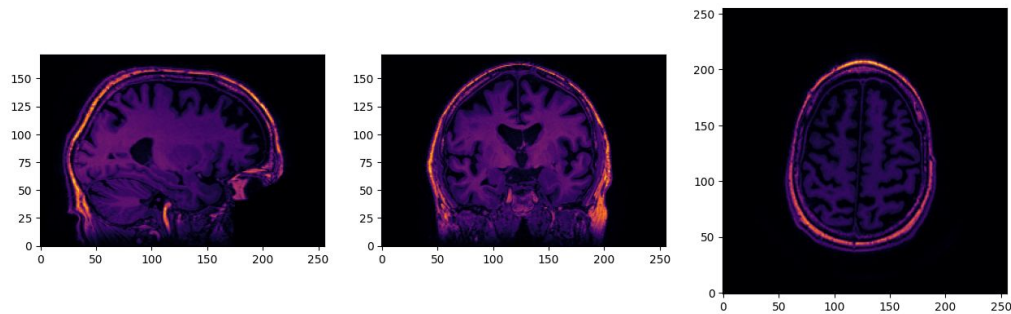
METHODS: Overview





METHODS: Data Collection

- Download 1772 T1-weighted MRI scans from Parkinson's Progressive Markers Institute (PPMI) [1]
 - 339 control
 - 926 prodromal
 - 504 PD



METHODS: Extracting Surfaces

- Extracting on Open Science Grid [22]
 - Job scheduler using HTCondor w/ ~80 participating institutions
 - Able to run thousands of jobs at once (526k jobs completed 11/16) [22]



METHODS: FreeSurfer Processing

- Aligns & transforms brain to known "atlas" using affine registration (determines difference between images & finds transformation parameters that minimizes differences)

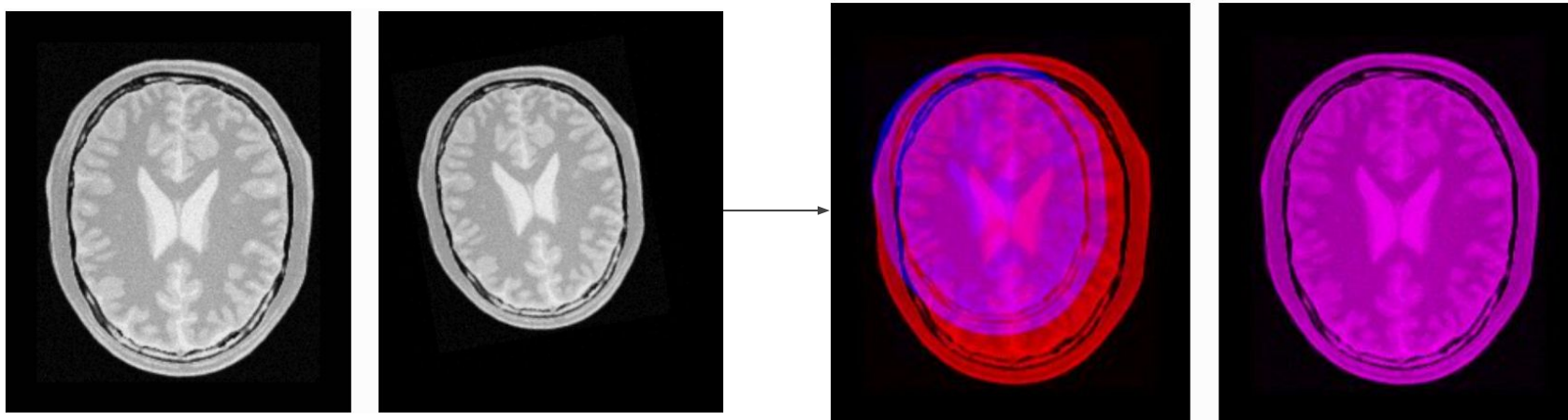
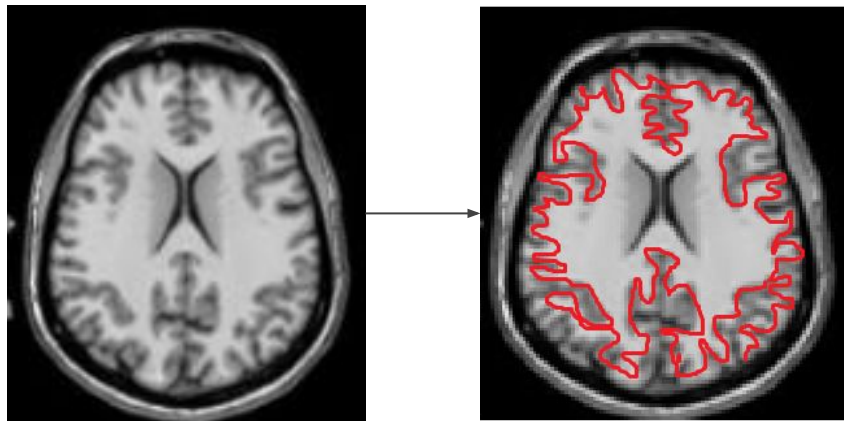


Image Source: [2]

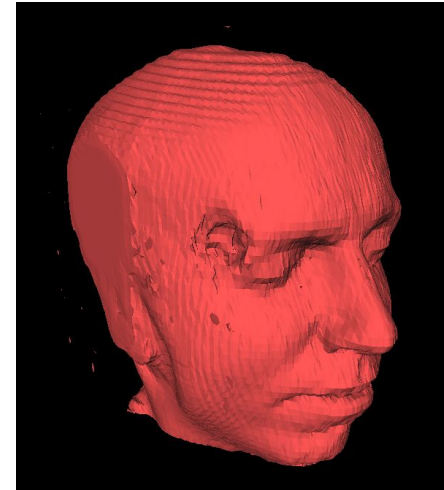
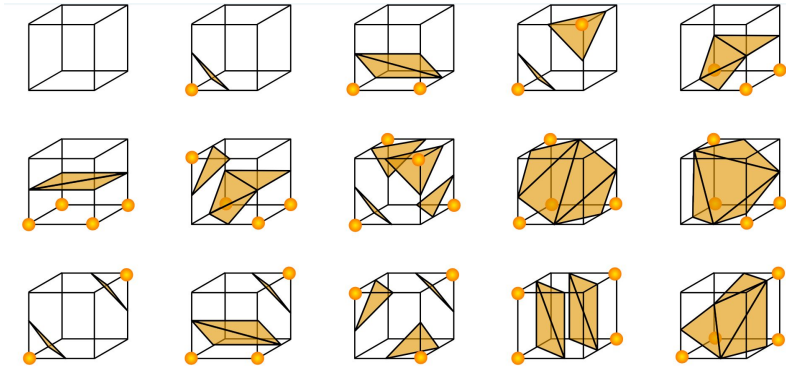
METHODS: FreeSurfer Processing

- Classifies points as white matter/gray matter based on intensity & probabilities (from "atlas")



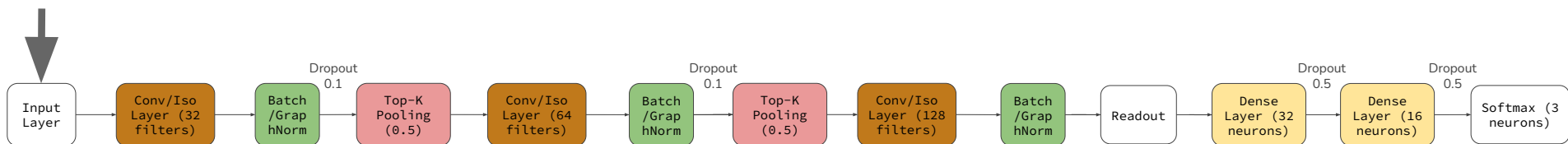
METHODS: FreeSurfer Processing

- Determines white matter surfaces w/ Marching Cubes; "marches" through voxels on boundary, determines vertices that meet intensity threshold; creates requisite polygon



METHODS: Extracting Graphs

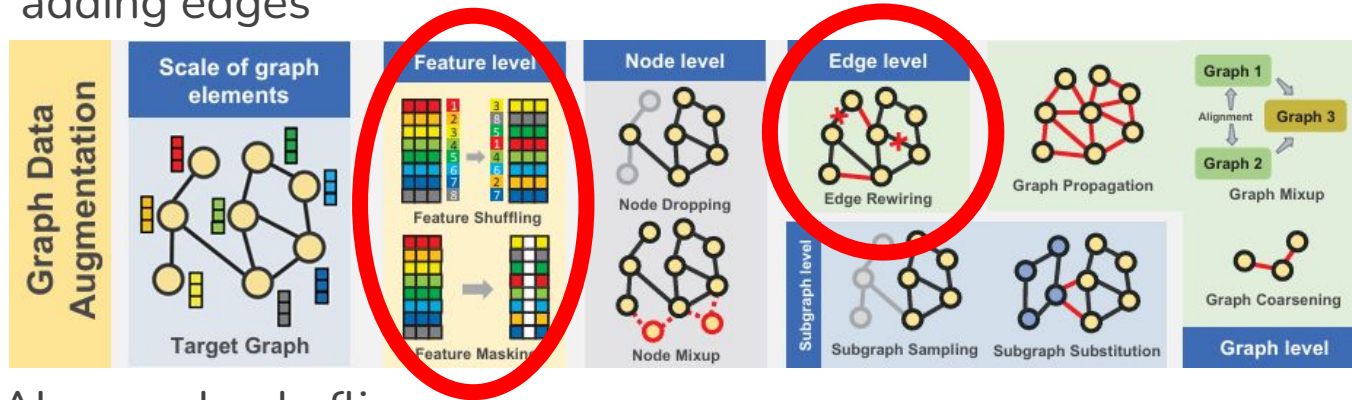
- Convert surfaces to graph representation w/ vertices as nodes (w/ features such as thickness, curvature, etc.)
- Vertices in same geometric "face" are connected with edges
- Feed surfaces into Graph Neural Network (Pytorch Geometric)



METHODS: Data Augmentation

- Data Augmentation: create new graphs from existing graphs via small random perturbations to features & randomly deleting and adding edges

Image Source: [21]



- Also randomly flip scans
- Can be used to rebalance classes & expand existing training set

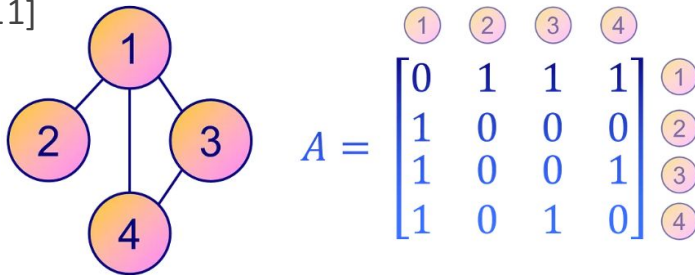


METHODS: After Augmentation

- 155 scans failed during extraction, proceeded with 1617 scans
 - 455 PD
 - 845 Prodromal
 - 314 Control
- After augmentation, 2792 instances
 - 910 PD
 - 940 Prodromal
 - 942 Control

For GCN (Graph Convolutional Network) Layers [27]

Image Source: [11]



Assume each node has 2 features

Feature matrix:

$$E = \begin{bmatrix} 3 & 4 \\ 1 & 2 \\ 5 & 0 \\ -1 & 6 \end{bmatrix}$$

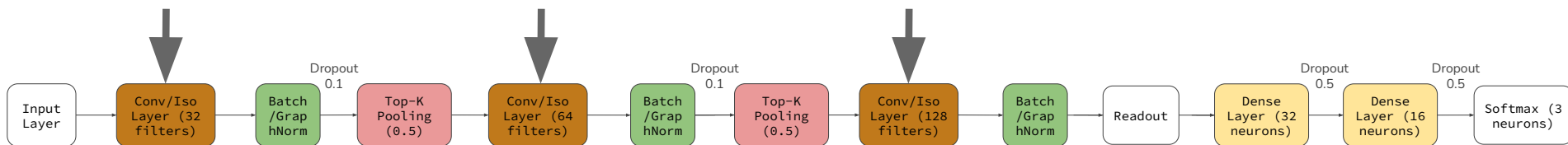
To get aggregated feature matrix: $E_a = E + AE$

$$E_a = \begin{bmatrix} 8 & 12 \\ 4 & 6 \\ 7 & 10 \\ 7 & 10 \end{bmatrix}$$

Then normalize according to # of neighbors: $E_a = (1/D)E_a$

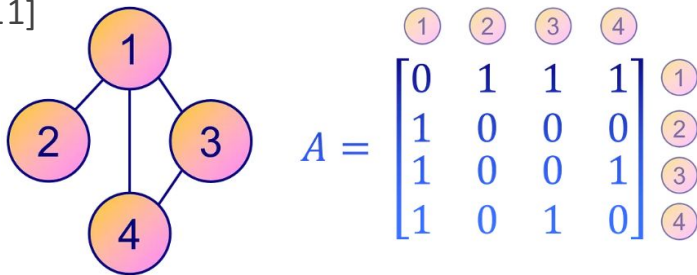
$$E_a = \begin{bmatrix} 8/3 & 12 \\ 4 & 6 \\ 7/2 & 5 \\ 7/2 & 5 \end{bmatrix}$$

Apply trainable weight matrix: $E_n = E_a W$
 $\sigma(E_n)$ is feature matrix for next layer



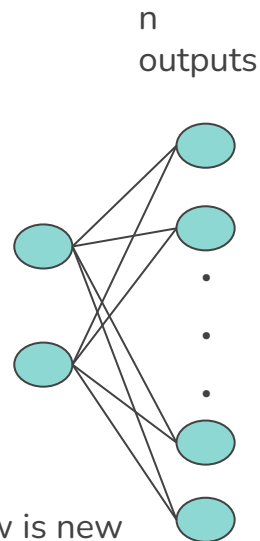
For GIN (Graph Isomorphism Network) Layers [28]

Image Source: [11]



Same up to aggregation of node w/ neighbors; then pass each node's features into 2-layer MLP (e.g. inputs $\rightarrow$ outputs); used ReLU activations

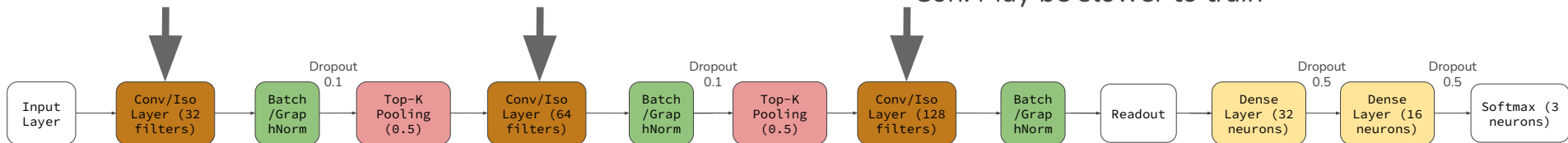
$$E_a = \begin{bmatrix} 8 & 12 \\ 4 & 6 \\ 7 & 10 \\ 7 & 10 \end{bmatrix}$$



Then $\sigma(E_n)$ (where each row is new representation of node from MLP) is passed into next layer

Pro: Better for graph-level classification; more expressive

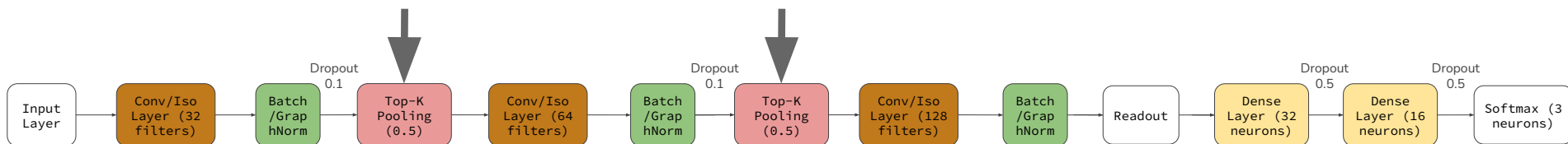
Con: May be slower to train





Pooling

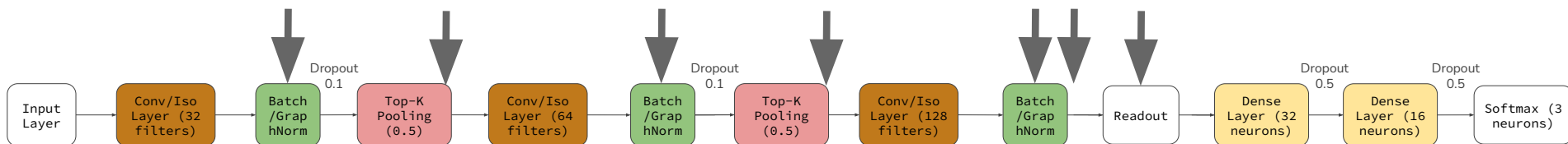
- Pooling in GCN: Top-k
- Each node is given score of $X * p / |p|$, wherein X is feature vector of node and p is learnable weight vector
- Then take top-k, wherein K is # or certain proportion; discard rest
- Scale retained nodes' features by $\tanh(\text{score})$





Readout & Normalization

- Batch Normalization z-score normalizes features across minibatch
- Graph Normalization z-score normalizes features across graph
- Readout extracts representation; concatenates mean & max of each feature across graph after each Graph-Norm-Pooling block & final graph
 - Then concatenate each of those & pass into FC layers





METHODS: Training Parameters

- 100 epochs w/ negative log-likelihood as loss function [5]
 - Likelihood: probability of correct class
 - Higher NLL -> worse prediction
- AdamW as optimizer; Adaptive learning rate & Weight decay [23]
 - β_1 , β_2 , and ϵ are all tunable hyperparameters; used 0.9, 0.999, and $1e-8$
 - α (learning rate) is (0.002), λ (weight decay rate) is (0.01); hyperparameters are similar to those used in [14]



RESULTS

W/ Convolutional layers & BatchNorm

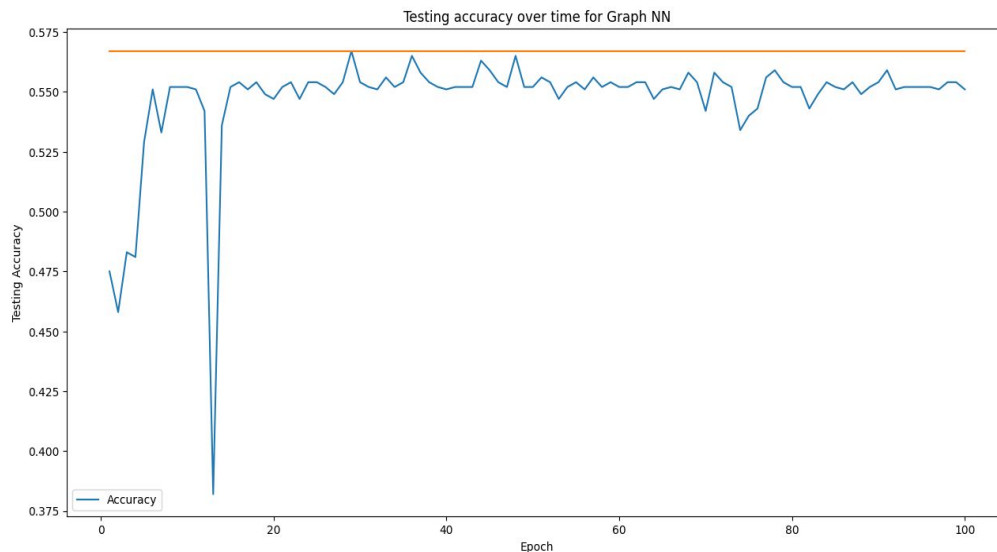
0.5671 testing acc

Pred→ Actual ↓	Control	Prodromal	PD
Control	151	38	0
Prodromal	22	166	0
PD	93	89	0

Recall control - 0.799

Recall prodromal - 0.883

0.5235 training acc





RESULTS

W/ Isomorphic layers & GraphNorm

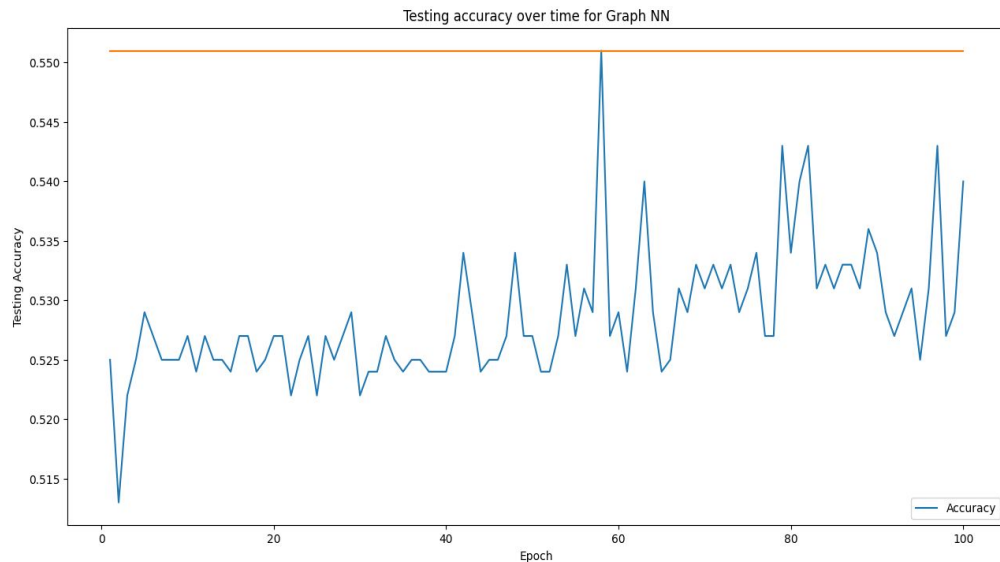
0.5510 testing acc

Pred→ Actual ↓	Control	Prodromal	PD
Control	113	61	15
Prodromal	14	167	7
PD	57	97	28

Recall Control - 0.672

Recall Prodromal - 0.888

0.5481 training acc





METHODS: Combined Models

- Apply Logistic Regression to Simpler dataset (patient symptom surveys)
- Merge w/ GCN to hopefully improve accuracy
- Preprocess: Merge 3 survey datasets (All numeric features)
- Impute missing value with MICE (Multiple Imputations through Chained Equations); applies estimator to predict missing values based on existing values (used Bayesian Ridge Regression)
- Restrict to only first 5 visits (prioritize early diagnosis)
- Drop date & ID information; rely on physical signals



RESULTS

Logistic Regression-1000 Iterations

- 0.859 training accuracy
- 0.818 training accuracy

Pred→ Actual ↓	Control	Prodromal	PD
Control	843	64	36
Prodromal	137	1938	22
PD	9	183	1169

Pred→ Actual ↓	Control	Prodromal	PD
Control	369	63	21
Prodromal	50	540	92
PD	8	44	344



COMBINED

- Used GCN & GraphNorm for Neural Network on Cortical Surfaces
- Used average of each patient's first five visits (as "augmented" graphs do not have corresponding visits) as input for Logistic Regression
- Combined predicted probabilities in length-6 vector
- Fed into MLP with structure 6 inputs, 9 hidden, 5 hidden, 3 outputs



RESULTS

W/ Convolutional layers & GraphNorm (normalizing across graph)

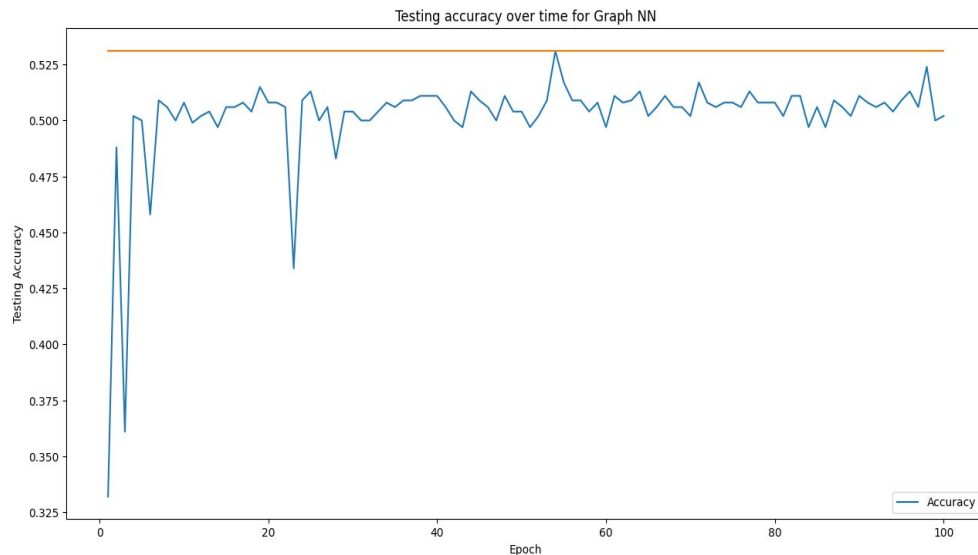
0.5313 testing acc

Pred→ Actual ↓	Control	Prodromal	PD
Control	119	63	7
Prodromal	28	159	1
PD	82	81	19

Recall control - 0.630

Recall prodromal - 0.846

0.5540 training acc





RESULTS

Combined Model (Logistic Regression + GCN/GraphNorm)

0.843 testing acc

Pred→ Actual ↓	Control	Prodromal	PD
Control	154	23	12
Prodromal	11	162	15
PD	7	20	155

Recall control - 0.815

Recall prodromal - 0.862

0.844 training acc



LIMITATIONS

- Not able to use GPU's on TJHSST cluster; prohibitively long training time
- Larger networks used in [8] & [14]; mine has only 26k parameters...
- [14] used VGG-16, DenseNet, InceptionV3, etc. which have between ~8 and ~138 Million parameters
- [8] used much larger custom CNN arch w/ ~41 million parameters



CONCLUSIONS

- GCN/GIN has some ability to differentiate different degrees of Parkinson's Disease
 - Graph-based NN was able to somewhat improve on the accuracy when combined with Logistic Regression model

FUTURE WORK

- Continue tuning GCN (dropout? Changing model size/biases?)
- Tune pre-trained 3D CNN (e.g. VGG 17) and compare against GCN performance



Q&A

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THANKS!